

Jeffrey Blay

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Professional Summary: Applied Science Ph.D. Candidate in Geospatial Data Science with 5+ years of experience in spatial AI, remote sensing, machine learning/computer vision, and geospatial data engineering. Builds scalable geospatial analytic solutions that transform geospatial datasets into actionable insights for evidence-based decision-making and address complex societal challenges.

Certifications

- IBM Professional Data Engineering - Coursera (In Progress – Expected, Summer 2026)
- Foundations of AI Engineering – CodePath (2026)

Education

North Carolina A&T State University, Ph.D. Applied Science and Technology (Expected December 2026)

- **Specialization:** Geospatial Data Science
- **Relevant Courses:** Machine Learning and Data Mining, Big Data Analytics, Advanced Geospatial Analysis, Neural networks, Statistical Methods, Multivariate Statistics for Engineers

Yale School of the Environment, Master of Environmental Science (May 2023)

- **Specialization:** Environmental Data science for Urban sustainability and resilience
- **Relevant Courses:** Advanced Remote Sensing, SDGs & Implementation, Statistics for Environmental Sciences, Data Exploration & Analysis, Environmental Data Science, Introductory Machine Learning

University Of Ghana, Bachelor of Arts Geography with Political Science (May 2019)

- **Concentration:** Geography with Political Science minor
- **GPA:** 3.81/4.0

Professional Experience

Geospatial Research Engineer

June 2026 – Present

Remote Sensing & GIS Lab, North Carolina A&T State University

- Building an open-source QGIS plugin that deploys CNN and transformer-based flood-depth models for on-demand inference by end users.
- Assessing disaster impacts on crop-specific land cover and vulnerable communities using federal datasets and Google Earth Engine.
- Leading a scientific publication on investigating physics-informed AI methodologies for flood prediction and environmental modeling.

Graduate Research Assistant, NASA Flood Project

September 2025 – May 2026

Remote Sensing & GIS Lab, North Carolina A&T State University

- Built and optimized transformer segmentation architectures (Swin-UNet, SegFormer) in PyTorch for urban flood-depth prediction from geospatial datasets.
- Developed a physics-informed AI framework with terrain-aware hydrostatic constraints, improving the physical consistency of flood modeling.
- Benchmarked physics-informed against conventional deep-learning models across four architectures, reducing surface-gradient violations by ~53%.

Geospatial Data Science Research Assistant, NASA Data Project

June - August 2025

NASA DEAP, North Carolina A&T State University

- Developed and evaluated CNN models (UNet, Attention UNet, UNet++) for flood-depth modeling using geospatial datasets.
- Published ezprocess, an open-source geospatial preprocessing library (PyPI/GitHub) that cut data-prep time by ~70% across ML workflows.
- Managed codes, experiments, and library version control using Git and GitHub, enabling collaborative development across research teams.

Graduate Research Assistant, NASA Data Project**September 2024 – May 2025***Remote Sensing & GIS Lab, North Carolina A&T State University*

- Built geospatial ETL pipelines using python (rasterio, GDAL) ingest, align, and preprocess ~347 million pixel-level records from multi-source raster data for training computer vision models.
- Led a 3-person team to design and publish a novel flood-depth benchmark dataset for reproducible flood inundation modeling.
- Implemented U-Net and pix2pix cGAN workflows to infer flood depth from UAV imagery and LiDAR point clouds.

Data Science Research Assistant, NSF Multimodal Data Fusion**June - August 2024***GEMS Institute, North Carolina A&T State University*

- Built ETL workflows using NumPy and scikit-learn to prepare 700k+ geospatial records for Random Forest and XGBoost flood prediction models.
- Collaborated with a 5-member team to develop and publish a multi-source geospatial dataset for flood inundation modeling; designed automated pipelines preprocess and integrate multi-sensor datasets using ArcPy in ArcGIS Pro.
- Completed training workshop in cloud-based data storage management and retrieval using Amazon S3.

Graduate Research Assistant, NOAA Flood Project**September 2023 – May 2024***Remote Sensing & GIS Lab, North Carolina A&T State University*

- Built python web-scraping workflows leveraging BeautifulSoup and SQLite to collect and curate custom image classification datasets for machine learning applications.
- Built a framework using PyTorch to train and evaluate image classification models (GoogleNet, ResNet18) for automated flood classification of crowdsourced images.
- Led a technical assessment of GAN-based approaches for flood mapping using remote sensing.

Community Group Manager**May – August 2023***Urban Resources Initiative*

- Supervised and managed 7 different community greenspace groups.
- Coordinated weekly project initiatives and strategies for fostering the green goals of each community group.
- Engaged with each community group to design and visualize technical ideas using ArcGIS Pro and QGIS.

Teaching Fellow**January 2023 – May 2023***Real-world Environmental Data Science Course, Yale University*

- Held office meetings to assist 12 students with python programming assignments, projects, and general class issues.
- Assisted the Class to utilize Google Collaboratory, GitHub, and GitHub classroom for teaching.

Research Assistant, NASA Environmental Justice Project**September 2022 – May 2023***Hixon Center For Urban Ecology, Yale University*

- Utilized multi-spectral satellite imagery and geostatistical techniques to conduct census block-level analysis to identify UHI-vulnerable urban communities.
- Conducted a block level analysis of Surface temperature using ArcPy and ArcGIS Pro SQL query builder.
- Used R (sp, dplyr, ggplot2) for time-series statistical analysis and mapping of urban heat and tree cover gaps.

Research Fellow, TRI Research Fellowship**May 2022 – May 2023***Tropical Research Institute (TRI), Yale University*

- Developed deep learning datasets and trained a UNet regression model for predicting building composition in metropolitan Ghana using satellite imagery.
- Leveraged Google Earth Engine to process VIIRS night-time light data for analyzing electricity distribution.
- Applied geostatistics to identify and map electricity infrastructure gaps in built-up areas.
- Designed and deployed a real-time data pipeline using ArcGIS Field Maps and Information Dashboards to collect and analyze 300+ ground truth building data, improving data collection accuracy by 70%.

Geospatial Research Assistant, Urban Africa Project**September 2021 - May 2022***Seto Lab, Yale University*

- Developed and optimized Google Earth Engine workflows for large-scale satellite data acquisition and preprocessing.
- Trained satellite image segmentation models in Python on a high-performance Linux computing cluster, implementing scalable batch processing pipelines.
- Built custom R and Python scripts for geospatial data processing, analysis, and visualizations.

GIS Data Officer

September 2020 - June 2021

National Census Secretariat, Ghana Statistical Service

- Performed spatial restructuring and quality assurance of Supervisory Area (SA) boundary maps, validating and correcting digitized census geospatial datasets.
- Collaborated with a 15-member Team to produce daily near real-time spatial coverage analytics to monitor enumerator field operations during national housing and population census activities.

Teaching / Research Assistant

September 2019 - August 2020

Remote Sensing/GIS Lab, University of Ghana

- Held Remote Sensing and GIS tutorial sessions for 150 undergraduate Remote Sensing/GIS students.
- Administered and Graded assignments and GIS group projects for the students.
- Assisted to undertake lab projects.

Physical Planning Intern

May - August 2018

Physical Planning Department, Awutu Senya East Municipal Authority

- Inspected land use activities within the assembly with Planning Officers.
- Used GIS techniques and Aerial images to analyze and coordinate land use activities within the Municipal Area.
- Built attribute data for land title registration and permit approval.

Publications

Blay, J., Gebregziabher, Y., Jha, M. K., & Hashemi Beni, L. (2026). Inundation2Depth: A multi-source dataset for floodwater depth estimation in urban areas. *Data in Brief*, 64, 112347. <https://doi.org/10.1016/j.dib.2025.112347>

Blay, J., & Hashemi-Beni, L. (2025a). Geospatial and Deep Learning Approaches for Modeling Floodwater Depth in Urbanized Areas. *Remote Sensing*, 18(1), 60. <https://doi.org/10.3390/rs18010060>

Blay, J., & Hashemi-Beni, L. (2025b). Pixels to Insights: Deep Learning for Floodwater Depth Mapping in Settlement Areas. *IGARSS 2025 - 2025 IEEE International Geoscience and Remote Sensing Symposium*, 1140–1144. <https://doi.org/10.1109/IGARSS55030.2025.11242913>

Blay, J., & Hashemi-Beni, L. (2025c, April). *Advanced Geo-Data Analytics and AI for 3D Flood Mapping to Protect Built Assets*. ISPRS Geospatial Week 2025, Dubai, UAE.

Blay, J., Fawakherji, M., & Hashemi-Beni, L. (2024, September 5). *Flood impact risk mapping in settlement areas from a 3D perspective: A case study of hurricane matthew*. 2024 IEEE International Geoscience and Remote Sensing Symposium, Athens, Greece. <https://doi.org/10.1109/IGARSS53475.2024.10640634>.

Blay, J., Seto, K. C., & Chen, T.-H. K. (2024). *Dark Development: Satellite Analysis of Building Density and Electricity Provision in Ghana's Urban Areas*. SSRN. <https://doi.org/10.2139/ssrn.5038955>

Fawakherji, M., Anokye, M., **Blay, J.**, & Hashemi-Beni, L. (2025). Multi-Resolution Data Fusion for Resilient Flood Mapping. *IEEE Access*, 13, 202275–202294. <https://doi.org/10.1109/ACCESS.2025.3636732>

Fawakherji, M., **Blay, J.**, Anokye, M., Hashemi-Beni, L., & Dorton, J. (2025). DeepFlood for Inundated Vegetation High-Resolution Dataset for Accurate Flood Mapping and Segmentation. *Scientific Data*, 12(1), 271. <https://doi.org/10.1038/s41597-025-04554-3>

Owusu, A. B., Mensah, C. A., **Blay, J.**, & Fynn, I. E. M. (2023). Urban Growth and Land Surface Temperature Dynamics: Lessons from Ghana. *Theoretical and Empirical Research in Urban Management*, 18(3).

Syum Gebre, T., Endale Ashebir, S., **Blay, J.**, Anokye, M., Pandey, V., & Hashemi-Beni, L. (2025). Real-Time Traffic Insights With Physics-Informed Neural Networks: Integrating the Aw-Rasele Model and LLMs. *IEEE Access*, 13, 181246–181266. <https://doi.org/10.1109/ACCESS.2025.3618282>

Conferences & Presentations

2025 IGARS (International Geoscience and Remote Sensing Symposium) Conference **August 7, 2025.**

- **Presentation Topic:** Pixels to Insights: Deep learning for Floodwater Depth Mapping in Settlement Areas.
- **Audience:** International

2025 ISPRS Geospatial Week **April 10, 2025.**

- **Presentation Topic:** Advanced Geo-Data Analytics and AI for 3D Flood Mapping to Protect Built Assets
- **Audience:** International

2025 Annual ASPRS and Geo Week Conference **February 12, 2025.**

- **Presentation Topic:** 3D Floodwater Depth Analytics with Big Geo-Data and cGAN.
- **Audience:** International

2024 Annual AGU (American Geophysical Union) Conference **December 11, 2024.**

- **Presentation Topic:** Urban Flood Depth Analytics: Leveraging Generative Algorithms and Remote Sensing for 3D Flood Depth Simulation.
- **Audience:** International

2024 NCAUG (North Carolina ArcGIS Users Group) Conference **September 24, 2024.**

- **Presentation Topic:** Leveraging AI and big geo-data for 3D flood depth mapping to safeguard-built assets.
- **Audience:** United States of America

2024 IGARS (International Geoscience and Remote Sensing Symposium) Conference **July 08, 2024.**

- **Presentation Topic:** Flood Impact Risk Mapping in Settlement Areas from a 3D Perspective: A Case Study of Hurricane Matthew.
- **Audience:** International

2024 Annual ASPRS and Geo Week Conference **February 14, 2024.**

- **Presentation Topic:** A 3D perspective of analysing the effect of Hurricane Matthew in small communities: A case study of Grifton, NC.
- **Audience:** International

2023 Annual AGU (American Geophysical Union) Conference **December 14, 2023.**

- **Presentation Topic:** Multi-channel deep learning-based architecture for flood detection and mapping
- **Audience:** International

Tropical Resource Institute(TRI) Symposium **April 21, 2023.**

- **Presentation Topic:** Local Scale analysis of Urban Intensity and Electricity infrastructures in Ghana with multi-sensor satellite images and machine learning techniques; A case study of Metropolitan cities.
- **Audience:** TRI Directors and Fellows, and all interested persons .

39th Yale School of the Environment(YSE) Research Day **April 14, 2023.**

- **Presentation Topic:** Local Scale analysis of Urban Intensity and Electricity infrastructures in Ghana with multi-sensor satellite images and machine learning techniques; A case study of Metropolitan cities
- **Audience:** The Entire Yale Community and all interested persons.

Yale School of the Environment(YSE) Confluence Talk Series **March 24, 2023.**

- **Presentation Topic:** Local Scale analysis of Urban Intensity and Electricity infrastructures in Ghana with multi-sensor satellite images and machine learning techniques; A case study of Metropolitan cities
- **Audience:** Yale School of the Environment Community and all interested persons.

Honors & Awards

Graduate Student Awards

2026

Outstanding Doctoral Graduate Research Assistant Award for the College of Science and Technology - NC A&T State University

Lidar Leader Award

2024

Best Poster presentation - 2024 Annual ASPRS & Geo Week Conference

Grants & Fellowships

Graduate Research Assistant Fellowship

2023 - 2026

North Carolina A&T State University

Teaching Fellowship

2023

Yale School of the Environment(YSE)

TRI Research Endowment Fellowship

2022

Tropical Resource Institute, Yale University

YSE need based and Merit Scholarship (Master's Study)

2021 – 2023

Yale School of the Environment (YSE)

Projects

Quickview Geodata Portal [Live](#) Python · FastAPI · GeoPandas · Railway Cloud· Leaflet.js · GitHub Pages

2026

- Built an open-source geospatial platform for fast, interactive spatial data exploration from URL for 5 file formats.

Weather Alert Dashboard [Live](#) Python · FastAPI · JavaScript · Docker · CI/CD · REST API · GitHub Pages

2026

- Built real-time weather monitoring platform integrating National Weather Service API; REST API with filtering, analytics, and risk-scoring across all 50 U.S. states.

Ezprocess - Open-Source Python Library Python · PyPI · GitHub · Geospatial ETL

2025

- Designed, packaged, and published geospatial data preprocessing library to PyPI/GitHub; reduces preparation time by 70% .

Technical Skills

Programming Languages: Python, R, SQL, JavaScript

Geospatial & Remote Sensing: ArcGIS Suite, QGIS, Google Earth Engine, ArcPy, GDAL, ENVI, SNAP

ML/ DL Frameworks: PyTorch, TensorFlow, Scikit-Learn, Computer Vision, Physics-Informed Neural Networks (PINNs)

Data Engineering & Processing: ETL Pipelines, Pandas, RDBMS, Web Scraping, Apache Airflow

Visualization & Decision Support: Matplotlib, Seaborn, Dashboards, ArcGIS StoryMaps, Power BI, R Shiny (R), Tableau

Web, APIs & Applications: FastAPI, RESTful APIs, HTML/CSS, Streamlit

Cloud, DevOps & Infrastructure: AWS S3, Docker, CI/CD, Git, GitHub, HPC, GitHub Pages, Render

Databases: PostgreSQL, MySQL, IBM DB2

Leadership and Co-Curricular Activity

Secretary & Social Media Manager

2024 – 2026

ASPRS Student Chapter, North Carolina A&T State University

- Managed the chapter's LinkedIn presence, creating and scheduling posts to increase engagement and visibility.
- Recorded meeting minutes and coordinated chapter activities, supporting event planning and organizational operations.

Lead instructor, Workshop on Geospatial Sciences

2025

CoST Young Scientist Day - North Carolina A&T State University

- Led a Geospatial Sciences workshop, mentoring high school students on career pathways.
- Demonstrated drones, GIS software, and 3D DTM technologies in hands-on sessions.

Student Representative

2022 – 2023

BIOMES Seminar Committee - Yale University

- Worked to invite speakers to speak at weekly Seminars and handle their logistics.
- Helped to facilitate students' engagement with speakers.

Co-Founder/Executive Director

2019 – Present

Good Lead Foundation

- Leads the organization to initiate and implement projects.
- Help raise funds to support educational activities within our community.
- Organize seasonal internal capacity-building workshops for members of the Foundation.